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Abdul Basit Fiaz

Electronics Engineer Fresh Graduate

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Driven Electrical Engineering fresh graduate with a strong background in electronics, embedded hardware design, RF/power systems exposure, and firmware programming. Skilled at taking a design from datasheet to working circuit, with hands-on experience across analog, digital, DSP, and control applications.

EXPERIENCE

Embedded Design Intern, NECOP Islamabad — Jun 2025 – Jul 2025

- Analyzed component datasheets to inform embedded system design decisions
- Designed PCB layouts for embedded hardware modules
- Developed STM32-based embedded hardware with cloud connectivity
- Implemented PID control algorithms to regulate DC motor speed

Sub Engineer E&M, MES Kharian Cantt — Jul 2024 – Aug 2024

- Assisted in operating and monitoring inverters, ACBs, transformers, and protection systems
- Monitored plant energy production through SCADA-based industrial control systems
- Studied data transmission protocols underlying SCADA network communication
- Proposed an IoT-SCADA integration solution to enable real-time monitoring and visibility of plant energy operations

EDUCATION

Bachelor of Electrical Engineering, National University of Sciences and Technology, Islamabad — Sep 2022 – June 2026
CGPA: 3.83 (Distinction: Silver Medal)

FINAL YEAR PROJECT

Embedded Glove for Sign Language to Speech Conversion

- Designed a sensor-based glove and implemented a real-time machine learning classification algorithm for gesture recognition
- Bridged a communication gap between the speech-impaired people and the society
- Generated a custom gesture-recognition dataset used to train the classification model
- Designed a novel optical linear encoder sensor, and glove system based on those sensors
- Developed custom PCBs of the System including ESP32,sensors, & BMS

ACADEMIC PROJECTS

20+ Electronics & Embedded Design Projects

- Designed and built 20+ electronics and embedded systems, cutting prototype development time by 30% across analog, digital, DSP, RF, and control applications
- Programmed ATmega328p, ESP32, STM32, PIC18F452, and Raspberry Pi, delivering reliable hardware control for several distinct projects
- Developed FPGA/Verilog digital hardware, implementing a 16-bit RISC processor that achieved 2× faster simulation speeds

Selected projects: 4-bit ALU (Arithmetic Logic Unit), Three-Channel DC Power Supply, Atmega32 & Ultrasonic-Based Toll Barrier System, Digital Ammeter, Audio Amplifier(Discrete), ECG Signal Filtering (DSP), EEG Filtering (DSP), DC Motor Speed Control (Discrete), FM Transmister & Receiver, Microstrip Tee Power Divider, Patch Antenna, DC-DC Closed-Loop Buck Converter using ESP32 as Controller, H-Bridge, ESP32-Based Home Automation, STM32 Cloud-Based DC Motor Control System, 16-bit RISC Microcontroller on FPGA.

CO-CURRICULAR AND LEADERSHIP

Head, Mentorship Wing, EE Department, CEME, NUST — Nov 2025 – May 2026

- Led departmental education-affairs initiatives, coordinating academic activities for the EE students
- Planned and managed technical and co-curricular events end-to-end, from concept through execution
- Represented the department in public-speaking engagements, strengthening communication and stakeholder coordination

ACHIEVEMENTS AND PUBLICATIONS

- Silver Medalist, CGPA 3.83/4.0
- Patent Filed (NUST) — Novel Optical Linear Encoder Sensor
- Commandant's Best Final Year Project of the Year at Open House.

CERTIFICATIONS

- Crash Course Python — Coursera
- Core MATLAB Skills — MATHWORKS
- PCB Designing Course — UDEMY x THETAY
- Wall of Hope Youth Leadership Program — University of Bolton x Tutify

- Machine Learning Specialization — Stanford University, Coursera

SKILLS

Circuit & PCB Design: LTSpice, Proteus, KiCAD, EasyEDA, PCB Layout, Design, Testing & Debugging
Embedded Systems: ATmega328p, ESP32, STM32, PIC18F452, Raspberry Pi (Linux), Embedded C: I2C, SPI, UART, Timers & Interrupts etc
RF & Power Systems: RF/Microwave Engineering, Antenna Design, SCADA, Power Systems, Protection Systems
Signal Processing & Control: Digital Signal Processing (DSP), PID Control, Control Systems, FPGA Design, Verilog/VHDL
Programming & Simulation: Python, MATLAB/Simulink, Machine Learning
Core & Soft Skills: Complex Problem Solving, Mathematical & Logical Thinking, Result-Oriented, Adaptability, Communication, Leadership, Project Management & Event Management

LANGUAGES

◇ Punjabi ◇ Urdu ◇ English